

Inside Large Language Models

Homework Assignment

Name:

Date:

Section 1: Matching (8 Items)

Match each LLM term with its definition. Write the letter in the box.

- A. Small chunks of text, like words, word-pieces, punctuation, or numbers, that an LLM reads and writes.
- B. The process of choosing the next likely token again and again to build an answer.
- C. The part of a transformer that helps the model focus on important earlier tokens in the context.
- D. A setting that controls how safe/predictable or random/creative the model's word choices are.
- E. Books, websites, code, and other examples the model studies to learn patterns before it is used.
- F. When an AI gives an answer that sounds confident but is made up or not supported by facts.
- G. The text already in the conversation that the model can use when predicting what comes next.
- H. A type of AI model trained on lots of language to predict and generate text.

- | | |
|----------------------|----------------------|
| <input type="text"/> | Tokens |
| <input type="text"/> | Next-word prediction |
| <input type="text"/> | Attention |
| <input type="text"/> | Temperature |
| <input type="text"/> | Training data |
| <input type="text"/> | Hallucination |
| <input type="text"/> | Context |
| <input type="text"/> | LLM |

Section 2: True or False (8 Questions)

Write T (True) or F (False) in each box.

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|----------------------|---|
| <input type="text"/> | A token is always exactly one full word. |
| <input type="text"/> | LLMs usually build answers one token at a time. |
| <input type="text"/> | Attention helps an LLM decide which earlier words or ideas matter for the next word. |
| <input type="text"/> | A higher temperature can make answers more surprising or creative. |
| <input type="text"/> | Training data teaches patterns, but it does not make the model know every fact perfectly. |
| <input type="text"/> | Hallucinations can happen when the model predicts likely-sounding text without reliable evidence. |
| <input type="text"/> | An LLM is the same thing as a live search engine. |
| <input type="text"/> | Clear prompts can give the model better context for its next-token predictions. |

Section 3: Short Answer (4 Questions)

Answer each question in 2-3 complete sentences.

1. Explain tokens using an example sentence. Why do tokens matter for how an LLM reads a prompt?

2. What does it mean that an LLM predicts the next word or token? How can this create a full answer?

3. Describe attention in your own words. What might the model pay attention to in the prompt: "The dog chased the ball because it was excited"?

4. Why can LLMs hallucinate? Name one thing you should do before trusting an important AI answer.

Section 4: Creative Challenge

Use your imagination! This is your chance to be creative.

Invent a tiny LLM helper for students. What is it called? What training data would help it? What temperature setting would you choose for facts vs. brainstorming? What rule would help it avoid hallucinations?

My LLM Helper Design:

Use the back of the page if you need more space!